

La Liga Project

Samuel Alfaro Ramirez | 09/14/2026

Business Task

Analyze historical La Liga match data to identify which factors are most associated with winning, in order to help a sports betting company set more accurate odds.

Stakeholders

Odds makers / trading team, risk analysts, and management.

Data Source

football-data.co.uk (PDDL license). 3 seasons, 1,140 matches.

Cleaning / Process

- Combined 3 tables from 3 seasons (23/24, 24/25, 25/26) into a single dataset.
- Added a "Season" column to keep them distinguishable.
- Removed the "Referee" column since it was empty.
- Verified zero missing values.

Analysis

- Loaded the dataset into BigQuery.
- Verified the load with a COUNT(*), confirming 1,140 matches.

Query 1

```
SELECT
  FTR,
  COUNT(*) AS count,
  ROUND(COUNT(*) * 100.0 / 1140, 1) AS percentage
FROM `laliga-proyect.Laliga.La Liga`
GROUP BY FTR
ORDER BY count DESC
```

Home advantage: the home team wins 45.8% of matches, compared to 28.2% for the away team and 26.1% draws (n=1,140, 3 seasons). Playing at home is a significant factor associated with winning.

Query 2

```
SELECT
  FTR,
  ROUND(AVG(HST), 2) AS avg_shots_on_target_home,
  ROUND(AVG(AST), 2) AS avg_shots_on_target_away
FROM `laliga-proyect.Laliga.La Liga`
GROUP BY FTR
ORDER BY FTR
```

Shots on target: the winning team averages more shots on target. The winning home team averages 5.70 vs 3.79 when it loses. The pattern holds on both sides (home and away). Strongly associated with winning.

Query 3

```
SELECT
  FTR,
  ROUND(AVG(HS), 2) AS avg_shots_home,
  ROUND(AVG(`AS`), 2) AS avg_shots_away
FROM `laliga-proyect.Laliga.La Liga`
GROUP BY FTR
ORDER BY FTR
```

Total shots vs shots on target: although both are associated with winning, shots on target discriminate better. When winning, the home team's total shots rise ~15% (12.56 → 14.39), but shots on target rise ~50% (3.79 → 5.70). Accuracy (quality) predicts winning better than volume (quantity).

Query 4

```
SELECT
  FTR,
  ROUND(AVG(HC), 2) AS avg_corners_home,
  ROUND(AVG(AC), 2) AS avg_corners_away
FROM `laliga-proyect.Laliga.La Liga`
GROUP BY FTR
ORDER BY FTR
```

Corners: NOT associated with winning. The averages are nearly flat across wins, draws, and losses (home: 5.3 when winning vs 5.67 when losing). Corners even tend to be higher for the losing team, possibly due to offensive pressure without effectiveness. Not recommended for predicting results.

Query 5

```
SELECT
  FTR,
  ROUND(AVG(HF), 2) AS avg_fouls_home,
  ROUND(AVG(AF), 2) AS avg_fouls_away
FROM `laliga-proyect.Laliga.La Liga`
GROUP BY FTR
ORDER BY FTR
```

Fouls: NOT associated with winning. The averages are nearly flat across results (12.3–13.0). The away team commits slightly more fouls than the home team in all cases, with no relation to who wins. Not recommended for predicting results.

Query 6

```
SELECT
  FTR,
  ROUND(AVG(HR), 2) AS avg_red_home,
  ROUND(AVG(AR), 2) AS avg_red_away,
  ROUND(AVG(HY), 2) AS avg_yellow_home,
  ROUND(AVG(AY), 2) AS avg_yellow_away
FROM `laliga-proyect.Laliga.La Liga`
GROUP BY FTR
ORDER BY FTR
```

Red cards: YES, associated with losing. The team with more red cards on average tends to lose (home and away), consistent with the disadvantage of playing a man down. Relevant negative factor. Yellow cards: weak effect, barely discriminate the result. The away team averages more yellows, consistent with committing more fouls.

Findings / Insights

Among the factors that predict a win, shots on target and home advantage were the most decisive in the analysis: the team with more shots on target has a higher chance of winning, and so does the home team. Total shots are associated with winning, but much less than shots on target, so accuracy matters more than volume.

Although they are infrequent, red cards are a clear negative factor: the team that receives one tends to lose.

As for corners, fouls, and yellow cards, these are factors that should not be taken into account when setting odds, since they do not discriminate between winners and losers.

In conclusion, the betting company should focus on which team is playing at home and which team tends to generate more shots on target. The other factors provide little value for predicting the result.

Share / Visualizations

I created a dashboard in Tableau Public with three visualizations summarizing the findings: (1) Home Advantage, showing the advantage of playing at home by result; (2) Shots on Target vs Match Outcome, comparing home and away shots on target by who wins; and (3) Which Match Stats Predict a Win, a ranking of which factors best discriminate between winning and losing.

Link: [View the dashboard on Tableau Public](#)

Act / Conclusion

Recommendation for the client (betting company): to set odds more accurately, focus on (1) home advantage and (2) each team's historical average of shots on target, which are the factors most associated with winning. Corners, fouls, and yellow cards do not add predictive value and should not weigh into the odds calculation.

Note: Red cards show a small absolute difference because they are rare events, but their relative impact is high — teams that receive one lose about 3x more often.